AUSTIN DONOVAN

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PROFESSIONAL SUMMARY

Data Analyst with 2+ years of professional experience and four end-to-end analytics projects spanning e-commerce performance, A/B testing, large-scale streaming data, and machine learning segmentation. Writes advanced SQL (CTEs, window functions), builds statistical and machine learning analyses in Python (Pandas, scikit-learn), and ships interactive Power BI dashboards with custom DAX data models. Experienced in ETL, data cleaning and validation, KPI reporting, and data storytelling, with a record of turning raw datasets of up to 26 million rows into statistically sound findings and clear business recommendations for stakeholders.

TECHNICAL SKILLS

Languages: SQL, Python, DAX, JavaScript, VBA

Databases: MySQL, Microsoft Access

Python Libraries: Pandas, scikit-learn, Matplotlib, Seaborn

Business Intelligence and Visualization: Power BI (data modeling, DAX, dashboard design), Excel (pivot tables, VLOOKUP), Google Sheets, KPI reporting, data visualization

Statistics and Machine Learning: A/B testing, hypothesis testing (z-test, chi-square), confidence intervals, bootstrap resampling, k-means clustering, RFM segmentation, exploratory data analysis (EDA)

Data Management: ETL workflows, data cleaning, data validation, data quality audits, data modeling, feature engineering, data governance

Tools and Platforms: Git, GitHub, Jupyter Notebook, Salesforce CRM

PROJECTS

Full case studies, SQL, notebooks, and dashboards at github.com/AustinLD

E-Commerce Performance and Customer Segmentation Analysis (Olist)

Analyzed 100,000+ orders from a Brazilian e-commerce marketplace to explain revenue trends, diagnose delivery-driven review problems, and identify the highest-value customer segments.

Technologies: MySQL, SQL (CTEs, window functions), Python (Pandas, Matplotlib, Seaborn), Power BI, DAX, Git, GitHub

- Designed a MySQL database across 9 relational tables and wrote advanced SQL queries with CTEs and window functions to analyze revenue, delivery performance, and customer behavior across 100,000+ orders.
- Quantified 20x monthly revenue growth (R$59K to R$1.19M) and showed it was volume-driven, with average order value flat between R$147 and R$182, including a 53% Black Friday spike.
- Identified late delivery as the clearest driver of poor reviews: on-time orders averaged a 4.29 review score versus 2.57 for late orders, a 1.72-point gap at a 91.9% on-time rate.
- Segmented 93,000+ customers with RFM analysis in Python, showing the Champions segment (6.6% of customers) generated R$1.72M in revenue while 97% of customers purchased only once, making repeat purchase the top growth lever.
- Built an interactive Power BI dashboard with custom DAX measures for revenue, delivery, and segment KPIs, and published a written case study with a reproducible GitHub repository.

Spotify Charts Analysis: Chart Dynamics and Streaming Concentration

Analyzed five years of daily US Top 200 chart data (7,878 songs, 2017-2021) to determine what separates lasting hits from one-week spikes and how concentrated streaming has become.

Technologies: Python (Pandas), SQL (window functions, CTEs), Power BI, DAX, Git, GitHub

- Processed a 26M-row (3.4GB) raw dataset in Pandas, filtering, validating, and reshaping it into a clean, analysis-ready regional slice.
- Modeled chart runs, rank deltas, and debut-versus-peak patterns in SQL, finding 55% of charting songs lasted a week or less (median run 6 days) and only 11% ever reached the top 10.
- Measured streaming concentration with SQL ranking analysis: the top 1% of songs captured 21% of all streams and the top 10% captured 73%.
- Attributed 9 of the 12 largest single-week climbs into the top 10 to external events rather than organic momentum, reframing the useful business signal from chart momentum to event detection.

- Designed a four-page Power BI dashboard on a custom dark theme covering streaming trends, chart dynamics, concentration, and top-10 climbers.

Marketing A/B Test: Ad Campaign Conversion Analysis

Evaluated a randomized ad-versus-PSA experiment covering 588,101 users to determine whether the ad campaign caused a real conversion lift and how large that lift was.

Technologies: Python (Pandas), SQL (CTEs, window functions), Power BI, DAX, Git, GitHub

- Proved a 43% relative conversion lift (2.55% versus 1.79%) with a two-proportion z-test ($z = 7.4$, $p < 0.001$) and a chi-square test, both rejecting equal conversion rates.
- Bounded the effect with a 95% confidence interval of 0.60 to 0.94 percentage points and a 10,000-sample bootstrap in which no resample fell at or below zero.
- Separated correlation from causation: conversion climbed from 0.16% (1 ad seen) to 17% (100+ ads), but exposure was not randomized, so the causal conclusion stayed on the randomized ad-versus-control gap.
- Wrote SQL analyses for conversion by group, exposure buckets, cumulative conversion, and day and hour segments, finding Monday converted best at 3.32%.
- Built a four-page Power BI experiment dashboard reporting lift, statistical significance, exposure, timing, and per-day treatment-versus-control lift.

Product Segmentation with K-Means Clustering (Online Retail II)

Clustered 4,479 SKUs from a UK retailer's 1.07M-row transaction log (GBP 19.9M revenue) into actionable product segments and benchmarked the machine learning result against a standard ABC/Pareto split.

Technologies: Python (Pandas, scikit-learn, Matplotlib), SQL (CTEs, window functions), Power BI, DAX, Git, GitHub

- Cleaned a 1.07M-row transaction log and engineered eight SKU-level features (sales velocity, price, seasonality, return rate, customer concentration), converting returns into a behavioral signal instead of discarding them.
- Applied log transforms and standardization to right-skewed features, then selected $k = 5$ clusters using elbow inertia and silhouette scores.
- Identified five product segments with distinct merchandising actions, including core bestsellers (24% of SKUs driving 78% of revenue) and a returns-prone segment with a 96% median return rate flagging quality issues.
- Benchmarked clustering against an ABC/Pareto revenue ranking in SQL, showing class A concealed 12 returns-prone and 38 single-buyer-dependent products that a flat ranking treats identically.
- Built a Power BI segmentation dashboard (cluster scatter, revenue treemap, segment profile table) and translated each segment into inventory, pricing, and promotion recommendations.

PROFESSIONAL EXPERIENCE

CWI Holdings | *Data Analyst* 2023 - 2025

- Designed, developed, and maintained a MySQL warehouse management database tracking 60+ material types, reducing theft and waste through tighter inventory control.
- Automated ETL and reporting workflows with Python, VBA, and Excel, cutting manual inventory update time and improving data accuracy.
- Built custom JavaScript functions for task automation and process improvement across internal data systems.
- Performed data quality audits covering null values, duplicates, format inconsistencies, and outliers, and built repeatable data cleansing workflows that reduced manual correction time.
- Applied data governance standards, keeping field definitions, naming conventions, metric logic, and access controls consistent across reports and dashboards.
- Delivered cross-department reporting and data visualizations in Excel and Google Sheets, and analyzed operational data to guide management decisions.

American Automobile Association (AAA) | *Client Specialist* 2022 - 2023

- Managed customer data records and supported reporting for client retention and membership analytics.
- Streamlined client onboarding and data entry with Salesforce CRM, reducing administrative errors and improving client retention.

EDUCATION

University of Central Florida | Full-Stack Web Development Coding Boot Camp (Full-Time) 2022

Full-Stack Web Development Certificate | [Verified credential (Parchment)]